Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

" To Eliminate the concern of floor plug missing & poor fit issue from Scorpio Model"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio –Xylo TCF.

	Plan	Actual
Start Date	**15.04.2019**	**20.04.2019**
Completion Date	**20.11.2019**	**20.11.2019**

Current Status: **Closed**

Cross Functional Team

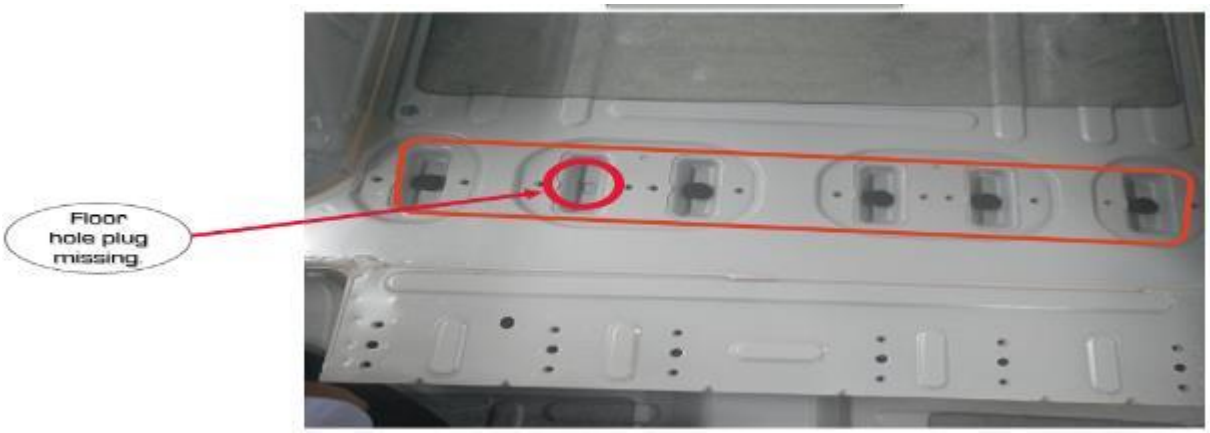
Sr. No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Prasad Deore	L7-0	STCF	9764005447
2	**Mentor**	Sandeep Chikhale	L5-M	STCF	9922408138
3	**Member 1**	SHIRWADKAR JAIDEEP	L7O	Paint PU	9890610151
4	**Member 2**	Satish Pagar	L8-O	STCF	9822209653
5	**Member 3**	Praveen Lakhe	L7-0	S body shop	02536608383
6	**Member 4**	Mahesh Savakar	L7-0	STCF	8888897833

Step1. Select Subject

Brief Description : To Eliminate the concern of floor plug missing & poor fit.

- What : Floor Plug Missing/ Poor Fit in Scorpio Vehicle.
- Where : At STCF assy Line Buy-off/CAI.
- When : Since Apr 2019.
- Who : By Buy off line inspector.
- Why : Repeat issue in Buy off,VI severity concern may leads to Water Leakage & possibility of dust insertion.
- How : Avg 5 vehicles per day.

NOT OK



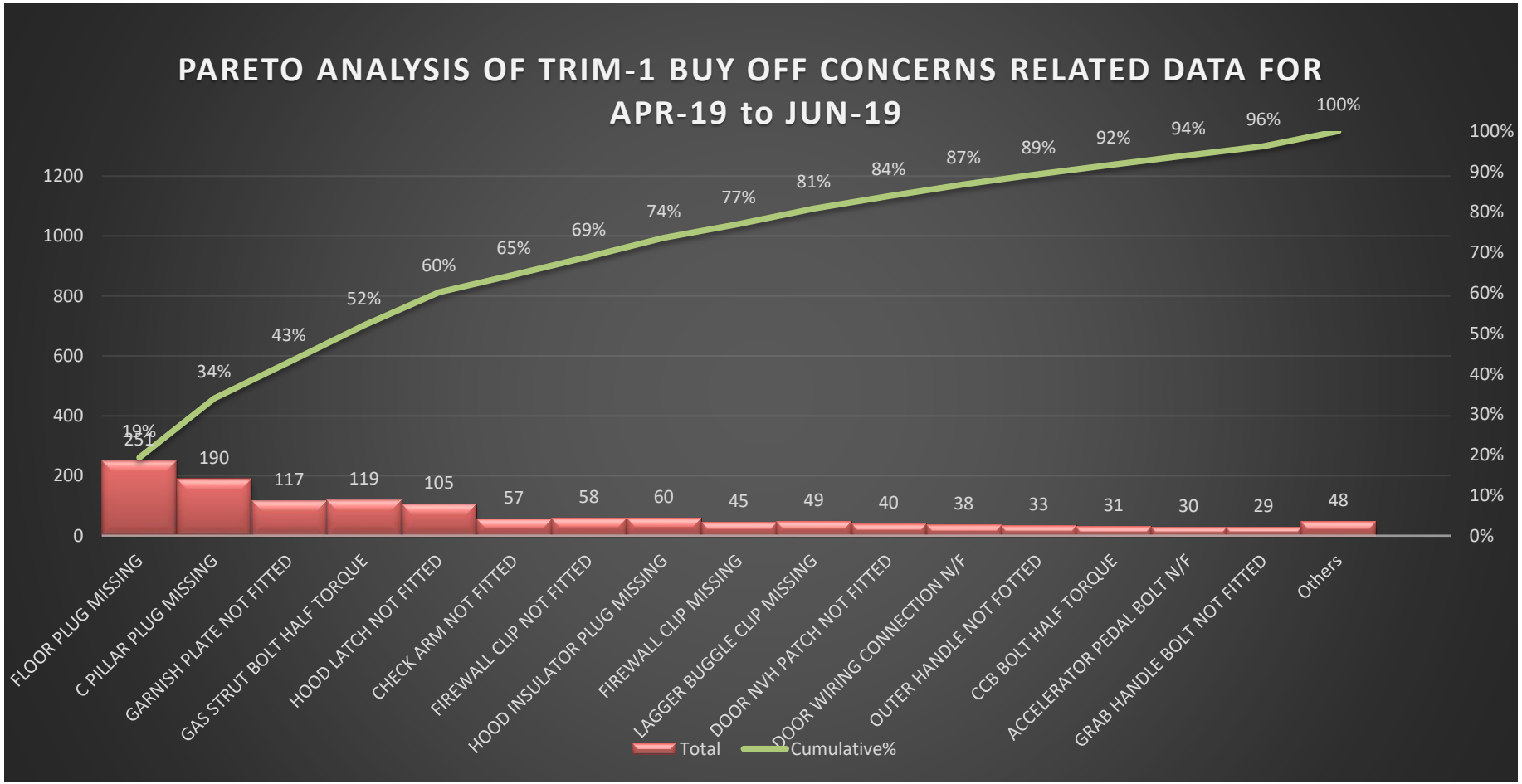
2. Justify the choice (reasons)

Average cost saved around Rs.7,20 per vehicle

Crusade 2.0

Departmental Objective :-
- FRC / FBO Improvement

2. Justify the choice



INFERENCE :-

AS PER PARETO GRAPH FLOOR PLUG MISSING IS TOP DEFECT AT TRIM-1 BUY OFF.

2. Justify the choice

Cost of Poor Quality-Rework

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	1
Avg Concerns/Day	Nos	5
Total Required Rework time per day	Min	5
Effective working time	Min	465
Operator required /day	nos	0.01
Approx Operator cost/ day	Rs	2256
Operator cost /concern	Rs	24.26
Avg no. of working days	days	25
Operator cost /month	Rs	606.45
Operator Cost/year	**Rs**	**7277.42**

2.b Immediate Containment Action

- ❑ Check point added at TRIM-1 buy off for floor panel plug missing checking.
- ❑ Informed to paint shop for removing excess sealer.

TRIM-1 BUY-OFF - CHECK POINTS

					Rev no:-167	Revision on - 16.03.18
Sr. N	SC	Description	What to Check	V1/V2	How to Check	Source of complaint
32		Lever of hood	Locking & clamping in bkt	V1	Visual	
33		Washer tube of 'A' pillar	Complete connection	V2	Visual	CAI
34		Wire routine of under driver & co driver seat	Clipping in holes & plug	V2	Visual	
35		Sunvisior plug	Check plug locked completely	V2	Visual & marking on plug	
36		Dog leg seal fitting	Correct fitment & fully clipping. Check cut & torn	V2	By hand	CAI
37		Plugging - floor & console area	All plug fitted on floor - As per model applicable matrix.	V2	Visual, marking for seat plug	ANOVAC
38		Plugging - C Pillar plug fitment	Complete locking of plug	V2	Visual & marking on body flange	RFI
39		Reg. Plate & logo fitment	Fitment as per model & no loose fitment	V1	By hand	CAI
40		Spacer fitting	Proper fitting	V2	Visual	
41		Tail door stopper	No loose fitment of screw/bolt. Foam in female stopper(Xylo)	V2	By hand & Marking	
42	CC	Tail door wire routing	check 5 nos. of clips inserted completely.	V2	visual	
43	SC/Z	Tail door latch connection	check connection & clipping.	V1	visual& Marking for clipping	LSP
44	CC	Registration lamp socket connection	Complete locking of socket	V1	By hand (Pull & push)	
45		"L' joint con. of rear washer tube	Complete insertion	V2	Marking	CAI
46		Fuel neck	No loose fitment of screw As per model - Silver passivation for 105 Silver passivation for U215	V2	by touching with hand and then do marking	105 U215

Step 3. Understanding Current Situation

Conformity Analysis
(L0 Analysis)

PU/ Area : Scorpio / Xylo TCF
Cell / Line : TRIM 1
BWT :01

Requested By : MQA | Date - 25/05/2019 | Source -Stage | Conf. Analysis Sr. [redacted]

Problem / Concern Statement : floor plug poor fit & missing.

FMEA Severity - Gravity : ☐ V1+ ☐ V1 ☑ V2 ☐ V3 | No of Incidences on the day (I) :20 | Vehicle/ Aggre. No. & model - Scorpio | Concern Priority Number :

Date of Incidence : 24/05/2019 | Shift / Time : Second | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

All In Process Vehicles Checked

Does a related Conformity Analysis or QC story already exist ? ☐ YES ☑ NO | If Yes: Conformity Analysis No / QC Story Sr. No

Failure mode or cause/s identified - Steering column mounting bracket on CCB- Stud slip

Problem Assigned to - Prasad Deore | BWT :02

Stage / Post Analysis -

Does a control already exist ? ☐ YES ☐ NO

Man (Human actions)

Operator name - Chetan Gangurde
Category - ☐ DR ☐ N1 ☑ EPP ☐ APP
Is operator trained on the stage ? ☑ YES ☐ NO
Operator level (I.L.U) & Dexterity level(1..4) | 1 | 2
Is operator under training ? ☐ under trg.on stage
☑ QC Stage ☐ Redeployed after 1 month ☐ 3*3 matrix completed
Bad rework ☐ YES ☑ NO | Newly deployed for rework - Yes
Is the specified facility used ? ☑ YES ☐ NO
Last calibration dates
Facility in good condition ? ☑ YES ☐ NO
Is it a substitute facility ? ☐ YES ☑ NO
Are machine parameters known ? ☑ YES ☐ NO
Parameters are respected ? ☑ YES ☐ NO

Method (Process)

Main steps are respected ? ☐ YES ☑ NO
Is there a related key point in SOP/CP/SOS ? ☑ YES ☐ NO
Are the reasons for key point known ? ☑ YES ☐ NO
Are operating reqmnts respected ? ☑ YES ☐ NO
Is the defect part of the checkman/ Buy off list ? ☐ YES ☑ NO
Was it detected by Inspector/ Buoff Opr ? ☐ YES ☑ NO
Buyoff opr.name - Sanjay Lone
In only one shift ? ☐ YES ☑ NO
In only one line ? ☑ YES ☐ NO
The problem follows the vehicle ? ☑ YES ☐ NO
The problem follows the part ? ☑ YES ☐ NO
Is it a repeat problem ? ☑ YES ☐ NO

Others

Is there a SOS/ SOP / CP/monitoring plan ? ☑ YES ☐ NO
Is it implemented ? ☑ YES ☐ NO
Does it conform ? ☑ YES ☐ NO
Is the packaging in good condition ? ☑ YES ☐ NO
Last prevetive measures taken on -
Is there any alert mechanism for failure ? ☐ YES ☑ NO
Is the specified product used ? ☑ YES ☐ NO
Did it evolve / Change ? ☐ YES ☐ NO
Is the storage SOP respected ? ☐ YES ☑ NO
Is the shelf life / expiry date respected ? ☐ YES ☑ NO

Machine | Product Failure | Material / Consumable

Note : 1. If both fields are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -
After analysis it was found that, stage associate not follow the SOS ,He has not pressed the plug & Buy off associate not detected the same at buy off stage.

Is Level 1 analysis required ? ☑ YES ☐ NO

1.Containment Action DRF given to all stage associate.	Resp	Date	Status
	PVD/ASP	25.05.2019	closed

2. Permanent Action - 1.	Resp	Date	Status

Sustenance Plan for PCA (Audit, Pokayoke)-
Analysed by Name/s - Prasad Deore | Date - 25.05.2019
Choice : ☐ WSS ☐ QC Story ☐ Human Error Checksheet

SPU/TCF/MQA/F/18

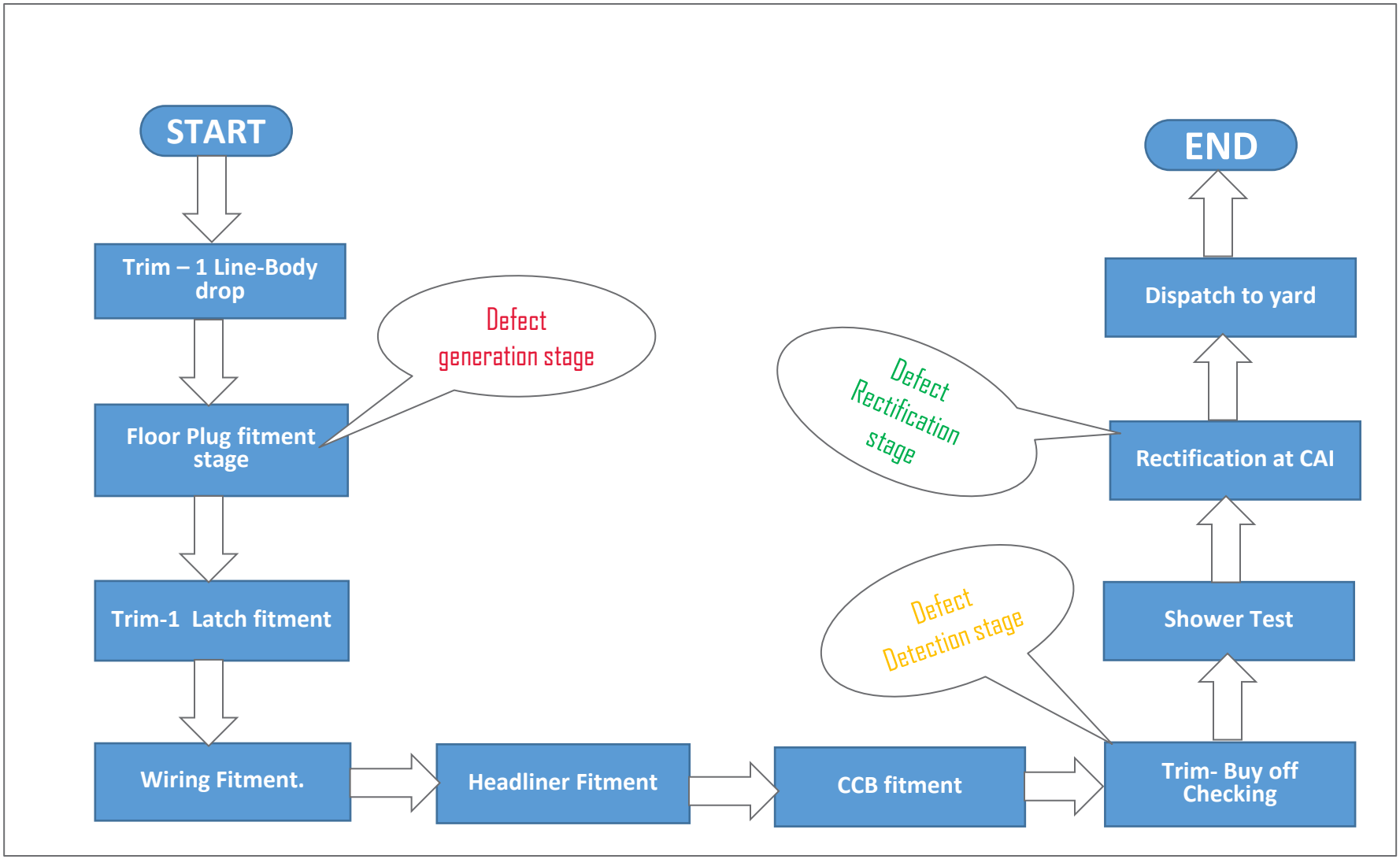
[redacted]

Inference from L0,
Problem follows with vehicle,
need detail analysis.

4M Analysis Good & Not so good observation	
Man	Operator Dexterial Level 3
Method	Operator aware about key point & check present at Buy off.
Product Failure	Problem follows Vehicle
Material	Handling & Shelf Life of Part adhered.

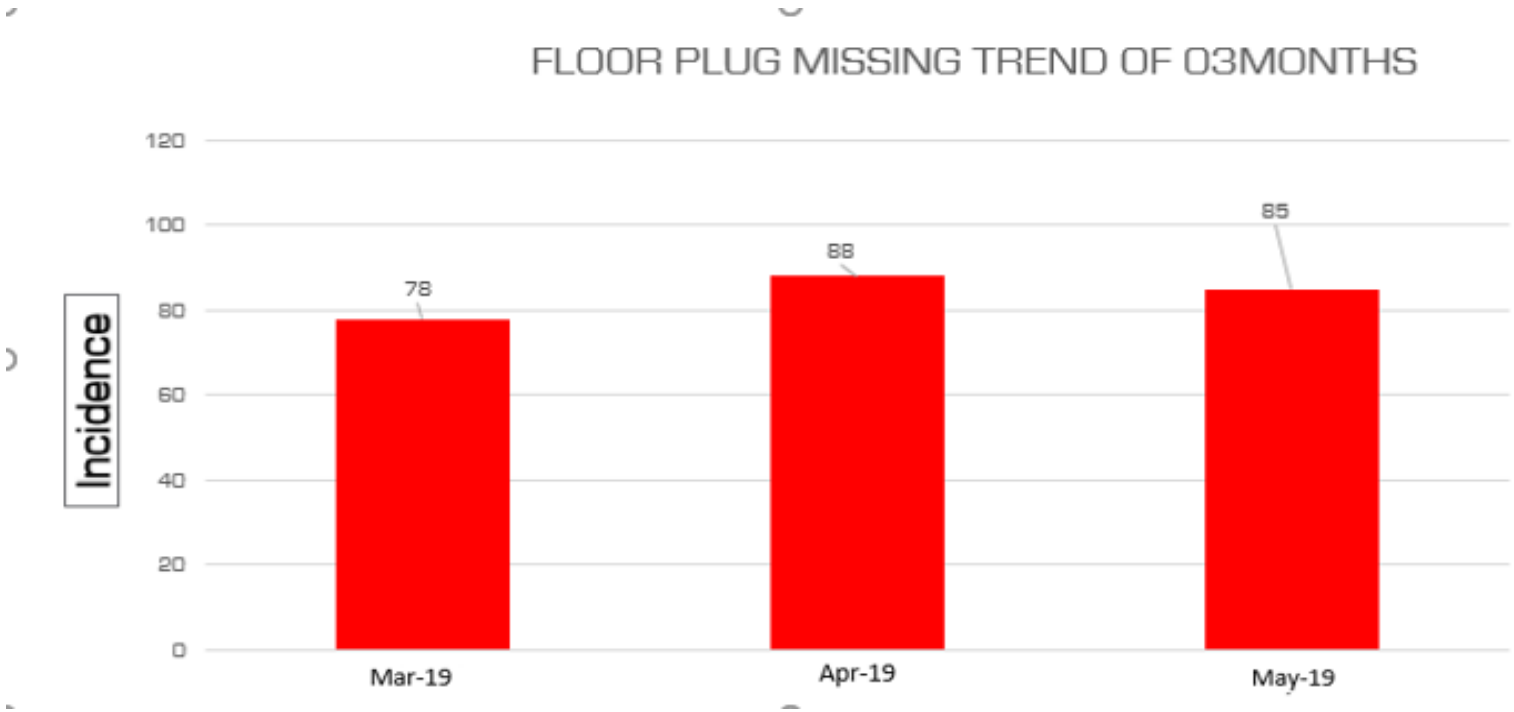
NO QUICK WINS

Step 3. Understanding Current Situation



3. Understand the current situation

Month wise Trend



INFERENCE :- As per graph Floor plug missing issue consistent month on month

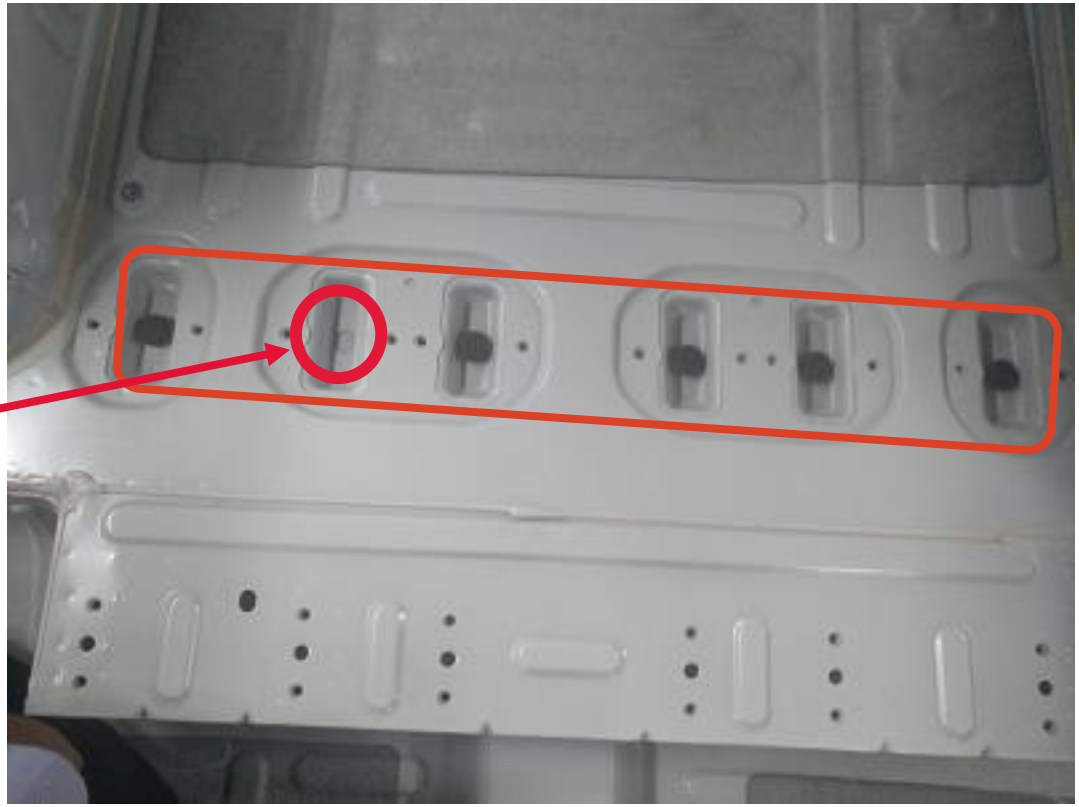
Step 3. Understanding Current Situation

Good & Not so good observations

- Issue observed in Scorpio all variant.
- Issue reported in both the shifts.
- Repair time is high around 1 minutes per vehicle.
- Issue reported at Trim-1 Buy off & CAI.
- There was level 3 operators for Floor Plugging stage.
- Not all the vehicles having this issue.
- Defect follows with Floor plugging.

Step 3. Understanding Current Situation

NOT OK

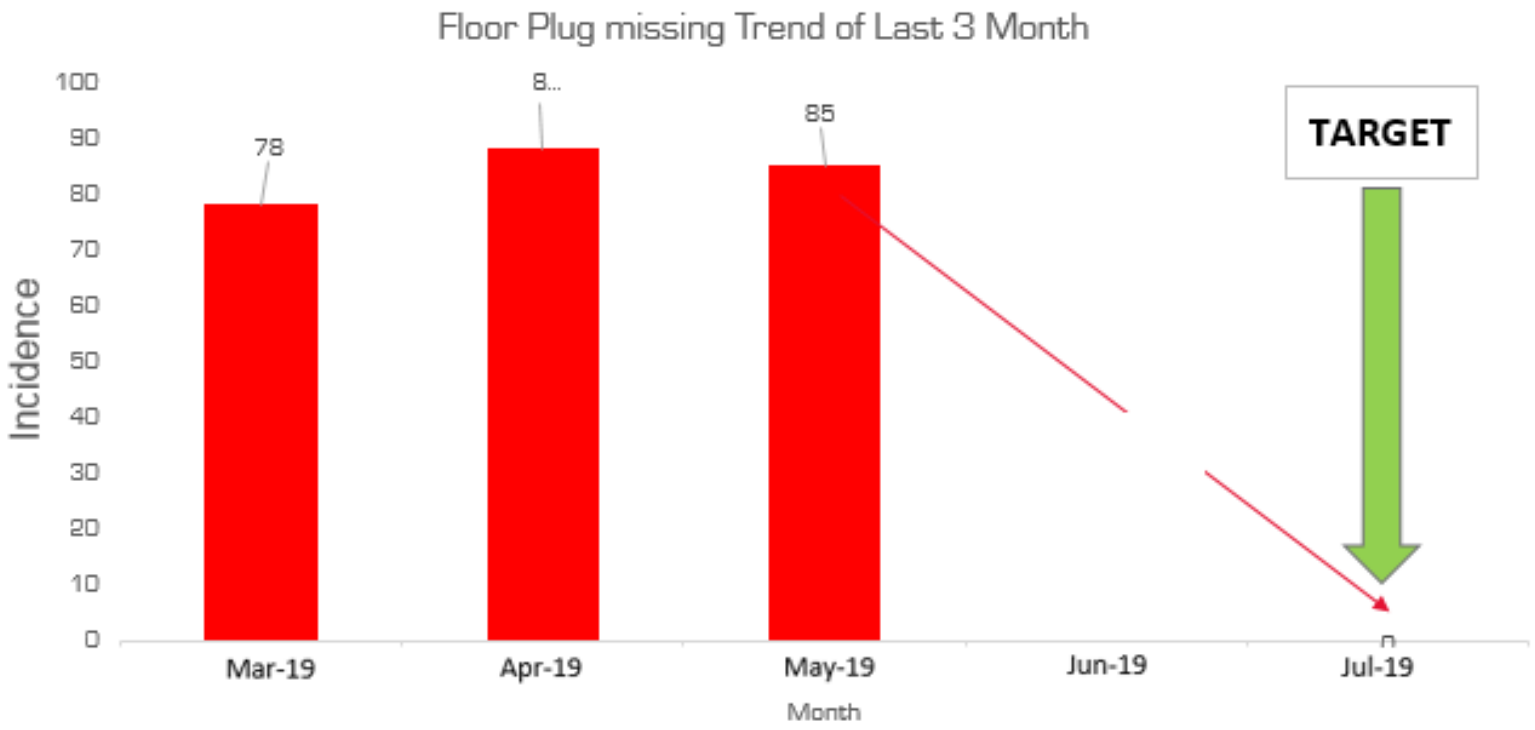


Floor hole plug missing.

Floor hole covered with Rubber plug/Grommet , possibility of leakage/ missing of plug.

4. Select targets

Present Level:	251 cases reported at Trim-Buyoff & CAI based on data of last 03 months
Target :	**Concern elimination by end of JUL-2019**



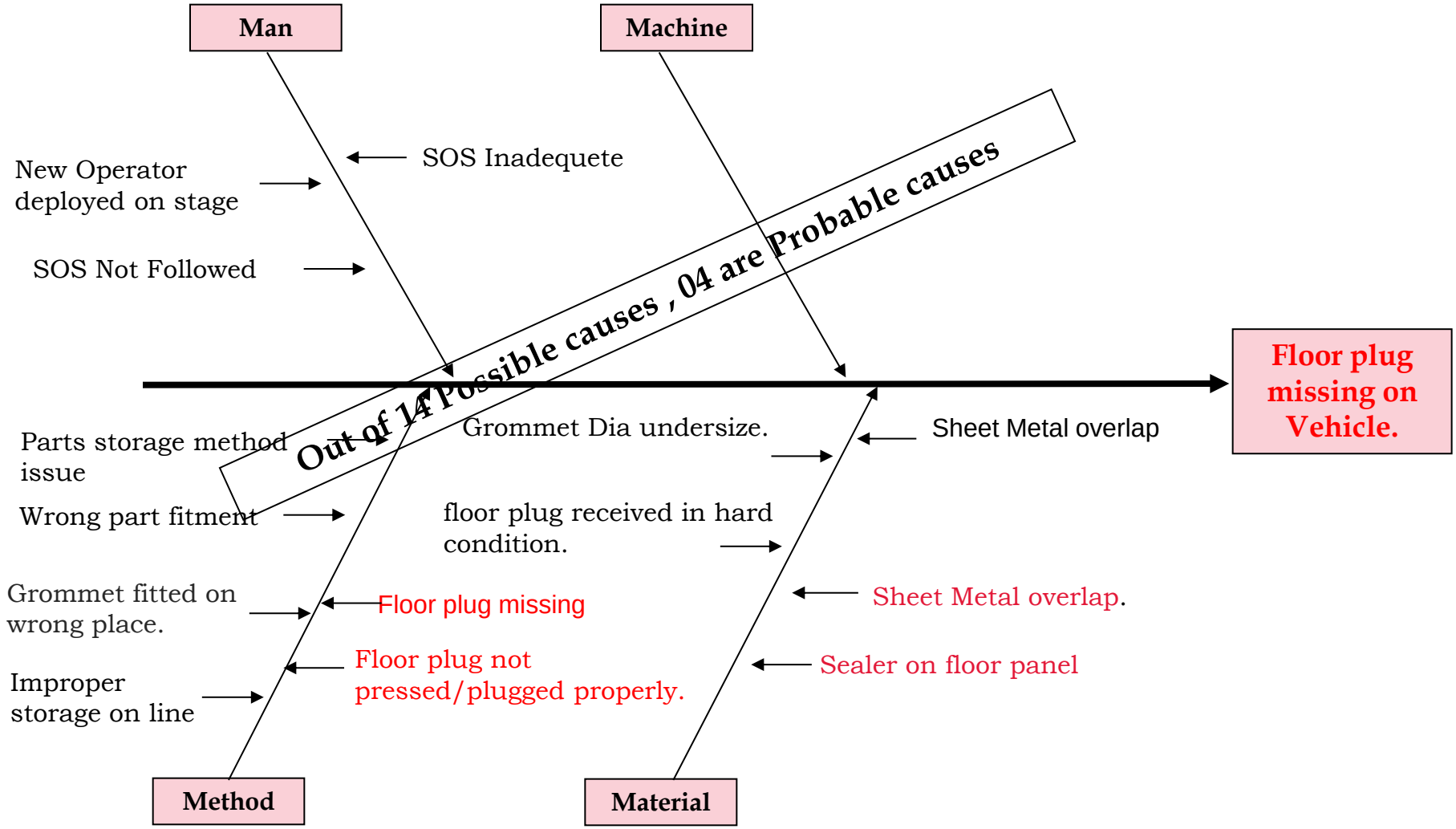
5. Analysis

List of Possible causes after brainstorming among team members

Sr No	Possible Cause
1	Floor plug not pressed/plugged properly.
2	Floor plug missing.
3	Sheet overlap.
4	Sealer on hole
5	Unskilled associate for floor plug fitment.
6	SOS Inadequate.
7	Floor panel damaged in received condition.
8	Sealer on floor panel.
9	Sheet Metal overlap
10	Grommet Dia undersize.
11	floor plug received in hard condition.
12	Wrong part fitment.
13	Grommet fitted on wrong place.
14	Improper storage on line
15	Improper packaging received from vendor.

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Possible causes identification in Brain storming session

Sr No	Possible Cause	Prasad	Satish	Jayesh	Mahesh	Praveen	Jaideep	Total
1	Floor plug not pressed/plugged properly.	9	9	9	9	3	9	48
2	Floor plug missing.	3	3	9	9	9	3	36
3	Sheet Metal overlap	9	9	9	9	3	9	48
4	Sealer on hole.	9	9	9	9	3	3	42
5	Unskilled associate for floor plug fitment.	1	1	3	3	3	3	14
6	SOS Inadequate.	3	9	9	1	1	1	24
7	Floor panel damaged in received condition.	3	3	1	1	1	3	12
8	Sealer on floor panel.	3	1	3	3	3	1	14
9	Grommet Dia undersize.	3	1	3	1	1	3	12
10	floor plug received in hard condition.	1	1	3	1	3	3	12
11	Wrong part fitment.	3	3	1	3	1	1	12
12	Grommet fitted on wrong place.	1	1	1	3	1	3	10
13	Improper storage on line	3	3	1	3	3	3	16
14	Improper packaging received from vendor.	1	3	1	1	3	1	10

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 36.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr. No	Not considered cause	Reason
5	Unskilled associate for floor plug fitment.	All trained associate deployed on stage
6	SOS Inadequate.	checked & found ok
7	Floor pannel damaged in received condition.	Checking started at body shop & found ok
10	Grommet Dia undersize.	Received stock checked found ok
11	floor plug received in hard condition.	Checked & found ok
12	Wrong part fitment.	correct part used.
13	Grommet fitted on wrong place.	Checked & found ok
14	Improper storage on line	Storage condition audited & found ok.
15	Improper packaging received from vendor.	checked & found ok as per PDS

Step 5. Analysis

List of causes considered as possible causes taken for test of hypothesis.

Sr No	Possible Cause
1	Floor plug not pressed/plugged properl
2	Floor plug missing.
3	Sheet Metal overlap.
4	Sealer on hole.

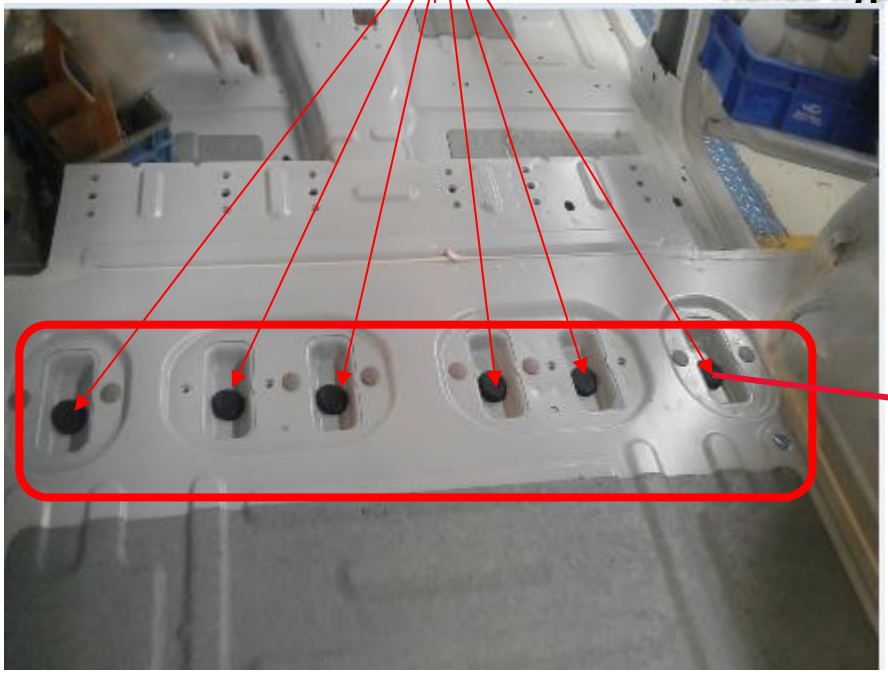
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Floor Plug Not plugged/Pressed properly.	Visual observation.	Checked & found that Rubber plug poorly fitted on vehicle.	**Hypothesis valid**

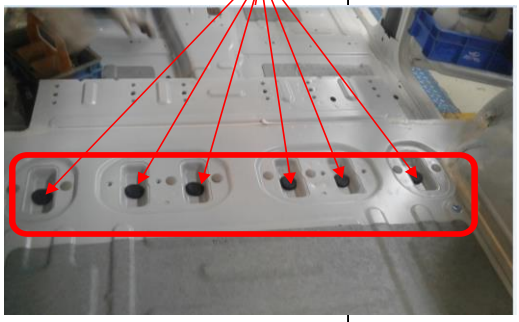
Part found that Rubber plug Not properly fitted on vehicle..

Hence hypothesis is valid.



CHECKSHEET

Desc	Floor Plug fitting	Date	19.05.2019	
Model	Scorpio			
Sr No	CHECK POINT	WHAT TO CHECK	HOW TO CHECK	Observation\Result
1	Floor Plug fitting	6 Grommet fitted on vehicle.	Visually & Marked	ok
2				ok
3				ok
4				ok
5				ok
6				ok
7				ok
8				ok
9				ok
10				ok
11				ok
12				Not Ok
13				Not Ok
14				Not Ok
15				Ok
16				Ok
17				Ok
18				Ok
19				Ok
20				Ok



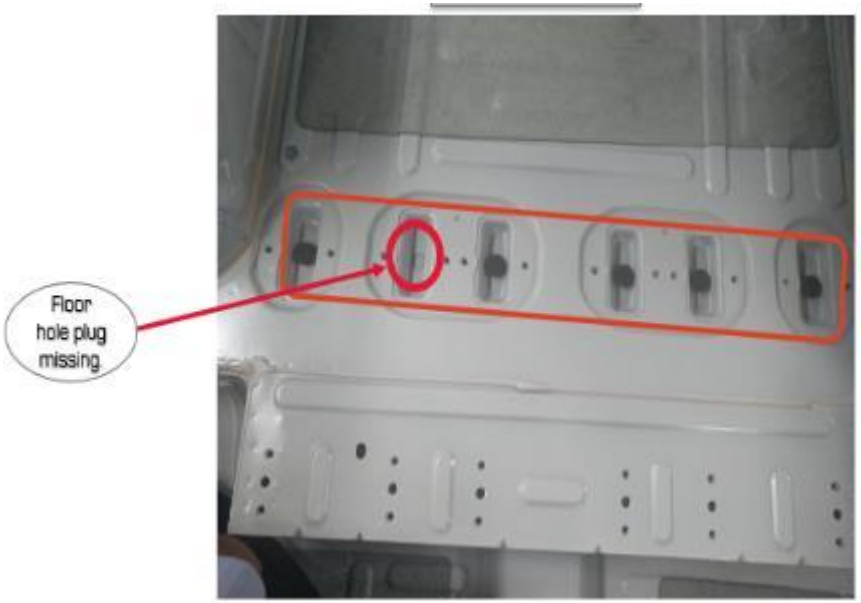
TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
2	Floor Plug Missing.	Visual Inspection at Buy off plug were missing/ not fitted on floor.	Its found that water coming inside the vehicle in shower test.	**Hypothesis valid**

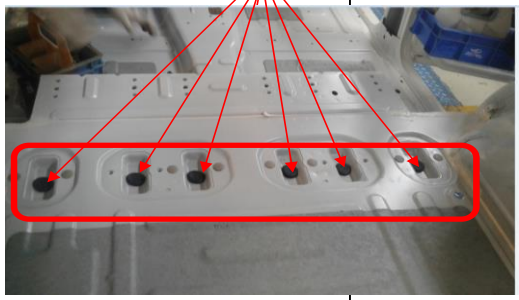
Its found that water coming inside the vehicle in shower test.

Hence hypothesis is valid.



CHECKSHEET

Desc	Floor Plug fitting	Date	19.05.2019	
Model	Scorpio			
Sr No	CHECK POINT	WHAT TO CHECK	HOW TO CHECK	Observation\Result
1	Floor Plug fitting	6 Grommet fitted on vehicle.	Visually & Marked	ok
2				ok
3				ok
4				ok
5				ok
6				ok
7				ok
8				ok
9				ok
10				ok
11				ok
12				Not Ok
13				Not Ok
14				Not Ok
15				Ok
16				Ok
17				Ok
18				Ok
19				Ok
20				Ok



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Sealer found on hole	Visually observed at plugging stage.	20 vehicle checked visually; all found are ok	**Hypothesis Invalid**

Checksheet				
Desc	Floor Plug	Date	13.05.2019	
Model	Scorpio			
Sr No	CHECK POINT	WHAT TO CHECK	HOW TO CHECK	Observatio
1	Sealer on Floor	Sealer on floor hole	Visually & Marked	ok
2				ok
3				ok
4				ok
5				ok
6				ok
7				ok
8				ok
9				ok
10				ok
11				ok
12				ok
13				ok
14				ok
15				ok
16				ok
17				ok
18				ok
19				ok
20				ok

20 vehicle checked visually in that Sealer are not found on hole.

Hence hypothesis is invalid.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
4	Sheet Metal overlap	Visually observed	20 vehicle checked; all found are ok.	**Hypothesis invalid**

Checksheet				
Desc	**Floor Plug**	**Date**	**13.05.2019**	
Model	**Scorpio**			
Sr No	**CHECK POINT**	**WHAT TO CHECK**	**HOW TO CHECK**	**Observatio**
1	Sheet Metal overlap.	Sheet metal overlap.	Visually & Marked	ok
2				ok
3				ok
4				ok
5				ok
6				ok
7				ok
8				ok
9				ok
10				ok
11				ok
12				ok
13				ok
14				ok
15				ok
16				ok
17				ok
18				ok
19				ok
20				ok

20 vehicle checked all found are ok.
Hence hypothesis is valid.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

Sr. No	Probable Cause	Testing method	Conclusion
1	Floor plug not pressed/plugged properly.	Visual Check	Hypothesis valid
2	Floor plug missing.	Visual check	Hypothesis valid
3	Sheet Metal overlap	Visual Check	Hypothesis invalid
4	Sealer on hole.	Visual check	Hypothesis invalid

2 Valid Causes

TEST OF HYPOTHESIS

5.3 Identify the root causes:

Sr No.	Probable cause	WHY?	WHY?	WHY?	Action
1	Floor plug not pressed/plugged properly.	For complete pressing of plug there are constraints	Due to Floor plug fitment is inside the cup shape area	Design constraints for plug pressing	Change the Design
2	Floor plug missing.	These plugs fitment is inside the cup area	Design constraints for visual detection	Associate forgot to fit one of plug	Change the Design / method

6. Implement the corrective measures

Possible Remedies Identified

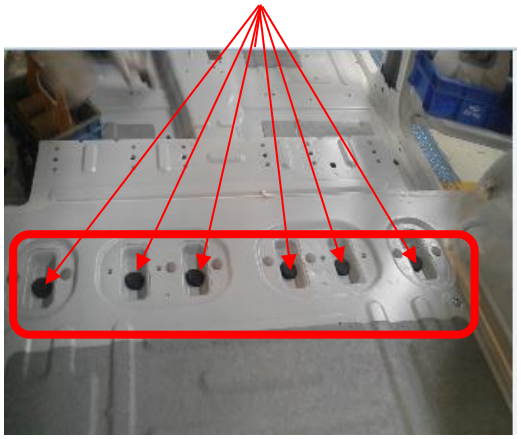
Sr. No.	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	Design constraints for plug pressing	Change the Design	Low	Low	Yes	Low	**Accepted**
2	Associate forgot to fit one of plug	Change the Design / method	High	High	No	Low	**Accepted**

6. Implement the corrective measures

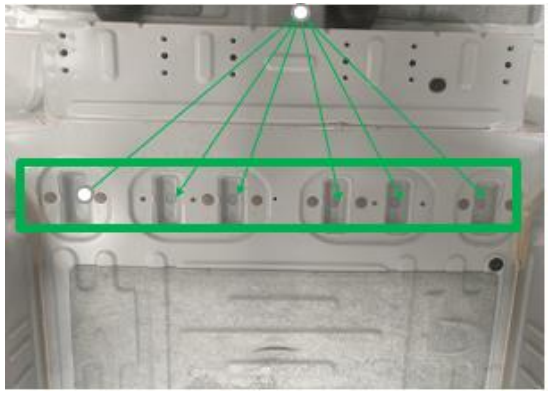
6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	Design constraints for plug pressing	Floor plug to be eliminate.	15.06.2019	Prasad/ Tushar Suryavanshi	Closed

Corrective Action : Floor plug to be eliminate from floor .



Before



After

- Since there was issue for floor plug fitment inside the cup , team decided to fit these plugs from underbody side.
- Underbody side plug fitment is possible at TCF due to accessibility issue.
- CFT decided to fit these parts at paint shop.
- During further analysis paint and R&D team has provided alternate plugs i.e Paintable Stickers which are less costly than rubber grommet
- CFT decided to go for paintable stickers instead of rubber grommets which is fitted at paint shop.

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
2	Floor plug missing.	Alternative solution to be develop for gromet.	17.06.2019	Prasad/Tushar Suryavanshi	Closed

Corrective Action : Deletion of these Rubber Plug(#0107FG0510N) Item No. 1 and use Paintable Sticker(#0107FU300140N) Item no. 2 at underbody at paint shop

Corrective Action :Deletion of these Rubber Plug(#0107FG0510N) Item No. 1 and use Paintable Sticker(#0107FU300140N) Item no. 2 at underbody at paint shop

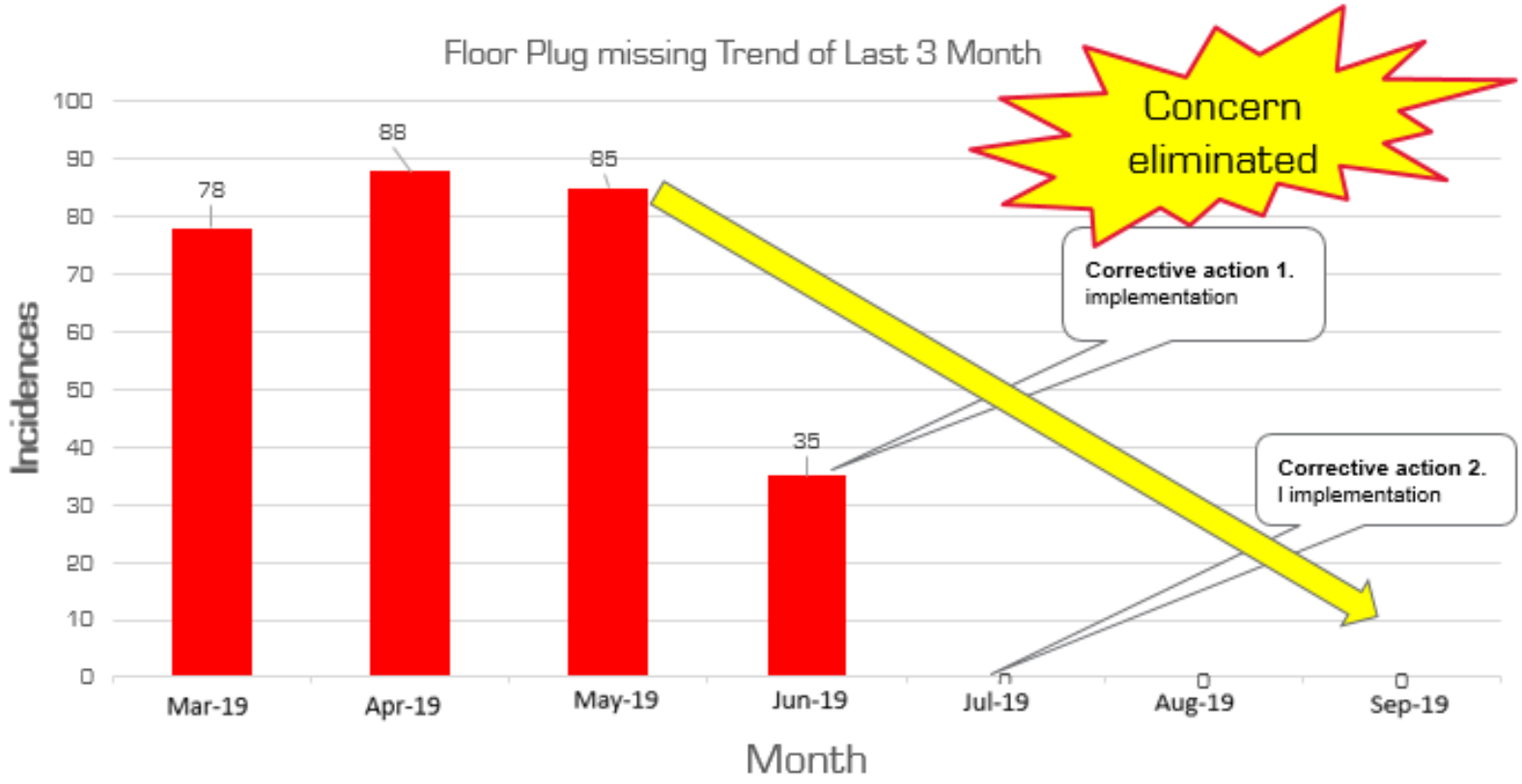
Item No. 1 -Existing part #0107FG0510N-Rubber Plug

Item No. 2 - Proposed plug of U-321:
0107FU300461N –Paintable sticker of 40 mm dia

1 .Feasible to use paintable sticker from bottom portion of floor at paint shop, and painting is possible on the same.

7. Check & confirm the results & effects

7.1 Verify Results:



INFERENCE :- As per graph issue reduced month on month after corrective actions.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S.N	Action Taken	Actual date	Responsibility	Status
1	Floor hole covered with adhesive paintable sticker.	15.06.2019	Prasad Deore	CLOSED
2	ECN update.& DCR sign off.	08.05.2019	Tushar Suryavanshi	Closed
3	FTR given to associate.	17.06.2019	Prasad Deore	CLOSED
4	OPL given to stage associate.	20.06.2019	Prasad Deore	CLOSED
5	Control Plan Updated	20.06.2019	Amar Pande	CLOSED
6	SOS Updation	20.06.2019	Prasad Deore	CLOSED

Check & confirm the results & effects (Cont.....)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) FRC increased.
2) Rework cost is reduced by approx.Rs.7277 /Annum.
3)Rs.7.20/per vehicle cost saved against replacement of grommet to sticker

In tangible Benefits :

1) Learning of CFT working approach
2) Gemba Culture is learned
3) Operator Morale Increased
4) Improved Customer Satisfaction.

From: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Sent: Monday, February 10, 2020 4:41:28 PM
To: DEORE PRASAD-NASHIK <DEORE.PRASAD2@mahindra.com>
Subject: RE: Sir Please approve

Okay Approved

From: DEORE PRASAD-NASHIK
Sent: Monday, February 10, 2020 3:23 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Subject: Sir Please approve

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	1
Avg Concerns/Day	Nos	5
Total Required Rework time per day	Min	5
Effective working time	Min	465
Operator required /day	nos	0.01
Approx Operator cost/ day	Rs	2256
Operator cost /concern	Rs	24.26
Avg no. of working days	days	25
Operator cost /month	Rs	606.45
Operator Cost/year	**Rs**	**7277.42**

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	Engineering change note	08.05.2019	**Suryawanshi Tushar N**	Closed
2	FTR	17.06.2019	**Prasad Deore**	Closed
3	DCR Sign off	12.04.2019	**Suryawanshi Tushar N**	Closed

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Awareness given through DRF.	20.06.2019	Prasad/jayesh	Closed

8.3 Additional Audit & check

1.	Check point added in buy off check sheet	25.06.2019	Vikrant Ghule	Closed

8. Standardization

8.1 Updation of documents: Engineering change note

Engineering change note : Engineering Notice No : W110-EN-A003-000180

DCRN

Mahindra & Mahindra Ltd.
Product Development

"Confidential Report"

Page No 1

Mahindra Rise.

Engineering Notice Release: W110-EN-A003-000180
AD Product Development
Mahindra & Mahindra LTD(A.S)

Created Date	08-May-2019
Initiated Date	12-May-2019

Engineering Notice No : W110-EN-A003-000180

Released Date : 29-May-2019 09:23

Reference Change Req No : DCR043380

Location : AD-NASIK

Implementation Location : M&M-Nasik

Engineering Notice Description : Deletion of Rubber Plug(#0107FG0510N) and use Paintable Sticker(#0107FU300140N) at underbody at paint shop.

Reason : CFM improvement on scorpio

Quantity : 500-(Trial and Regular)

Reason Code : VAVE (Cost reduction / IMCR)

Priority : Desirable

Note : 20/05/2019

Affected Models : W110 SUV all varints,

Is EN Re Released? :

Parent EN NO :

Life Cycle State : Released

Owner Name : Suryawanshi Tushar N (301282)

Engineer : Suryawanshi Tushar N (301282)

Approved By : 207930-DEEPAK CHAUDHARI

Authorized By : VE-SCORPIO-HEAD_207094-GAUBA MANOJ

Legends

Add New : A Part/Assy that is getting released through ECN for the very first time

Modified : A Part/Assy that is already released through an ECN and has undergone some changes

Added with Modification : It is very similar to "Modified" status except that it is being added newly into a different Parent

Existing : A Part/Assy that is already being used is carried forward as it is to some other parent

Deleted : A Part/Assy that is being used is removed from a Parent

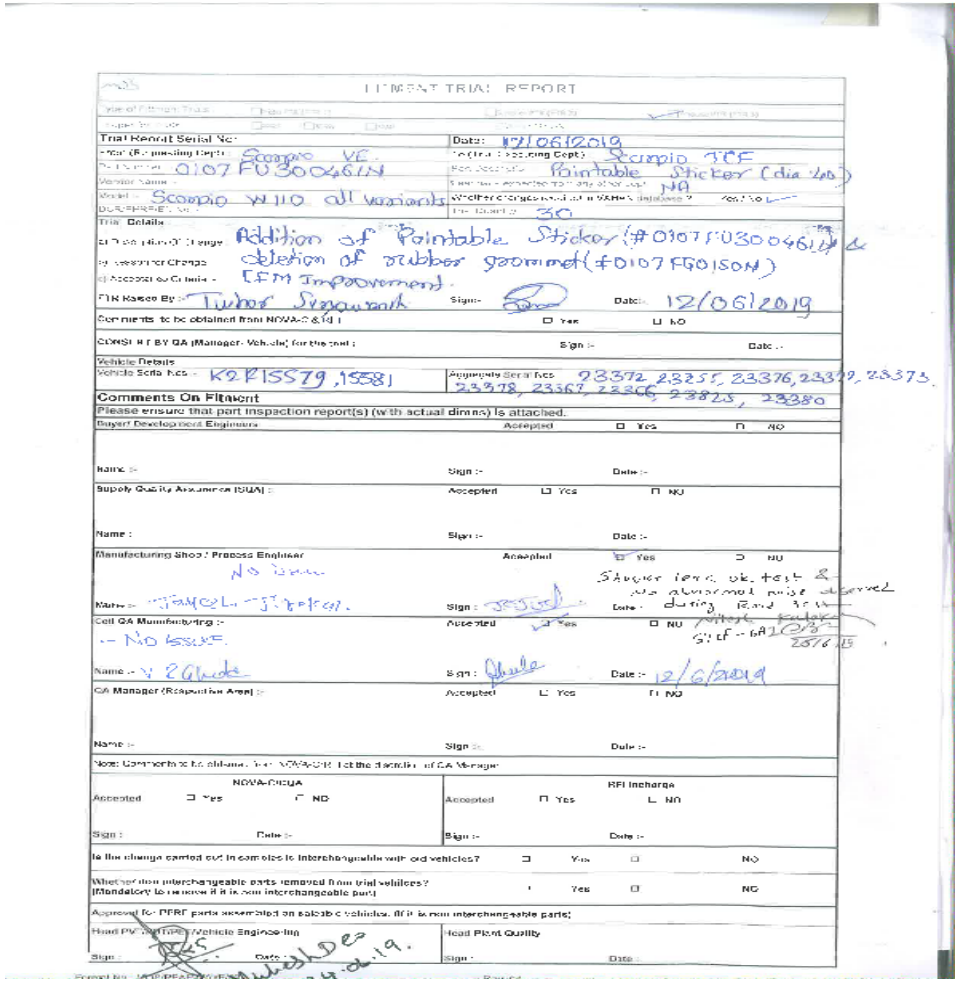
Reinstated : Reintroduction of a Part/Assy that was used Previously in M&M but was discontinued

ICA :
O - New part number can be used in place of old part no. & vice versa;
I - New part number can be used in place of old part no. but not vice versa;
Z - Old part number can be used in place of new part no. but not vice versa;
X - No ICA between new and old part numbers;
Y - Interchangeability as set of parts (to use new part no. we have to use more than one new part no);
NA - Not Applicable;

8. Standardization

8.1 Updation of documents:FTR

FTR



FITMENT TRIAL REPORT

Trial Report Serial No:

Date: 12/06/2019

Scorpio VC

Scorpio VCF

0107FU300461N

Pointable Sticker (dia 20)

Scorpio W110 all variants

NA

30

Trial Details

Addition of Pointable Sticker (#0107FU300461N) & deletion of rubber grommet (#0107FG015ON)

EFM Improvement.

FTR Raised By :- Tushar Singanwad

Sign:-

Date:- 12/06/2019

Comments to be obtained from NOVA-C & R I

Yes

NO

CONSIDERED BY QA (Manager- Vehicle) for the part :

Sign :-

Date :-

Vehicle Details

Vehicle Sr.No :- K2R15579, 15581

Aggregate Serial Nos 23372, 23355, 23376, 23379, 23373, 23378, 23367, 23366, 23825, 23380

Comments On Fitment

Please ensure that part inspection report(s) (with actual dimns) is attached.

Buyer Development Engineers

Accepted

Yes

NO

Name :-

Sign :-

Date :-

Supply Quality Assurance (SQA) :

Accepted

Yes

NO

Name :

Sign :-

Date :-

Manufacturing Shop / Process Engineer

Accepted

Yes

NO

No issue

Shaker lever ok test & no abnormal noise observed during road test

Name :- JAMGAL - J. Rokal.

Sign :

Date :

Cell QA Manufacturing :-

Accepted

Yes

NO

-- No issue.

Name :- Y. Ghude

Sign :

Date :- 12/6/2019

QA Manager (Respective Area) :-

Accepted

Yes

NO

Name :-

Sign :-

Date :-

Note: Comments to be obtained from NOVA-C&R for the decision of QA Manager

NOVA-C&QA

Accepted

Yes

NO

Sign :

Date :-

RFI Incharge

Accepted

Yes

NO

Sign :-

Date :-

Is the change carried out in same size is interchangeable with old vehicles?

Yes

NO

Whether non interchangeable parts removed from trial vehicles? (Mandatory to remove if it is non interchangeable part)

Yes

NO

Approval for PFRT parts accumulated on saleable vehicles. (If it is non interchangeable parts)

Head PV Manufacturing/Vehicle Engineering

Sign :

Date :

Head Plant Quality

Sign :

Date :

8. Standardization

8.1 Updation of document: DCR

DCR

Mahindra & Mahindra Ltd.

Mahindra Rise.

DESIGN CHANGE REQUEST

DCR Number	DX1804580
Source Of Change Request	207004 - GAUBA MANOJ
Product Owner	207004 - GAUDA MANOJ
Sub Product Owner	207930 - DEEPAK CHAUDHARI
Location	AD-NASIK
Reason Code	VAVI (Cost reduction / IMCR)
Change Description	Replacement of rubber grommet with paintable sticker on underbody near middle seat mtg area to avoid air leakage
Why this change ?	to avoid air leakage from middle seat area.
Proposed Change	Replacement of rubber grommet with paintable sticker on underbody near middle seat mtg area to avoid air leakage
Priority	
CAE Required	
CAE	
DFMEA/QFD Required	
Designer	
Lifecycle Status	Proposed

Estimated Cost Impact

Type	Saving	Increase
Material	7	
Conversion		
Man Power		

Affected Model/Major Agg List

Scorpio W110 all variants

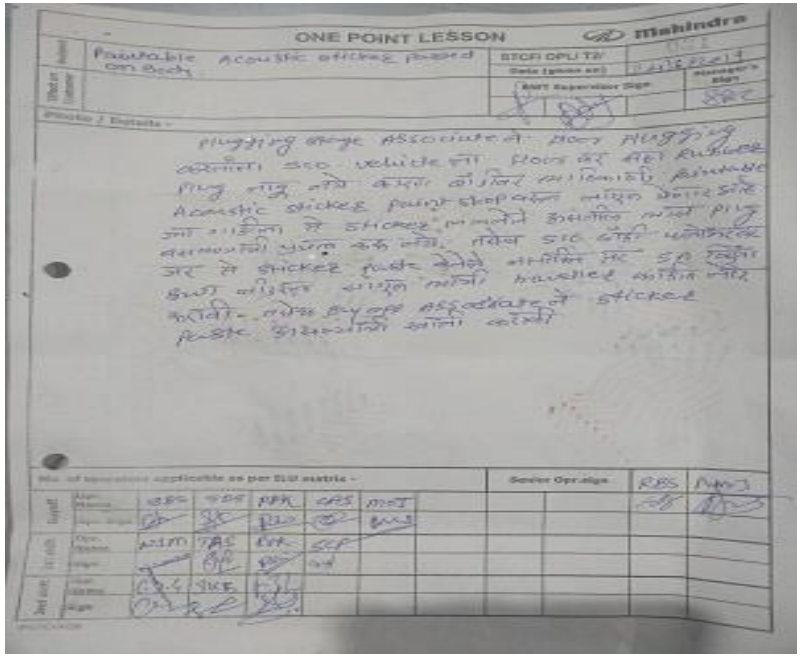
Approval Statge	Approver	Role	Status	Approval Date	Remarks
Initiation	301282 - SURYAWANSHI TUSHAR N	Initiator	Initiated	12-APR-2019	
Preliminary Approval by SPO	207930 - DEEPAK CHAUDHARI	SPO	In Progress		

Apr 12, 2019, 11:53 AM

8. Standardization

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Awareness given through OPL & SOS.	20.06.2019	Prasad/Jayesh	Closed



9. Summarize & Plan future actions

9.1 Conclusion:

Floor panel plug missing concern eliminated with finding alternative solution as plug is replaced with Paintable sticker of 40 mm dia.

9.2 Horizontal Deployment:

Analysis is useful in upcoming projects, inputs to DFMEA.

9.3 Lessons Learnt:

Things Gone Wrong :

Initially only mfg process was suspected as cause of concern. Unawareness of associate were not considered for floor plugging stage.

Things Gone Right :

Brainstorming Happened between CFT team & problem solved by QC story approach by **alternative thinking**. Alternate Solution were identified & developed & saved cost of organisation.

Thank you

Visit us at www.mahindra.com

Disclaimer

Mahindra & Mahindra herein referred to as M&M, and its subsidiary companies provide a wide array of presentations and reports, with the contributions of various professionals. These presentations and reports are for informational purposes and private circulation only and do not constitute an offer to buy or sell any securities mentioned therein. They do not purport to be a complete description of the markets conditions or developments referred to in the material. While utmost care has been taken in preparing the above, we claim no responsibility for their accuracy. We shall not be liable for any direct or indirect losses arising from the use thereof and the viewers are requested to use the information contained herein at their own risk. These presentations and reports should not be reproduced, re-circulated, published in any media, website or otherwise, in any form or manner, in part or as a whole, without the express consent in writing of M&M or its subsidiaries. Any unauthorized use, disclosure or public dissemination of information contained herein is prohibited. Unless specifically noted, M&M or any of its subsidiary companies is not responsible for the content of these presentations and/or the opinions of the presenters. Individual situations and local practices and standards may vary, so viewers and others utilizing information contained within a presentation are free to adopt differing standards and approaches as they see fit. You may not repackage or sell the presentation. Products and names mentioned in materials or presentations are the property of their respective owners and the mention of them does not constitute an endorsement by M&M or its subsidiary companies. Information contained in a presentation hosted or promoted by M&M is provided "as is" without warranty of any kind, either expressed or implied, including any warranty of merchantability or fitness for a particular purpose. M&M or its subsidiary companies assume no liability or responsibility for the contents of a presentation or the opinions expressed by the presenters. All expressions of opinion are subject to change without notice.

Copyright © 2012 Mahindra & Mahindra Ltd. All rights reserved.